

Linear Algebra - CS - 3A & 3B

Problem Sheet: Change of Bases, Column, Row and Null Spaces

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A Bases and change of bases

1. (**Mechanical**) Show that the set $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ is a basis for $\mathbb{R}^2$. Express $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ in this basis.
2. (**Mechanical**) In $\mathbb{R}^2$, let $\mathcal{B} = \{(1, 2), (2, 1)\}$ and $\mathcal{C} = \{(1, 0), (0, 1)\}$. Find the change-of-basis matrix from $\mathcal{B}$ to $\mathcal{C}$ and from $\mathcal{C}$ to $\mathcal{B}$.
3. (**Conceptual**) If $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a basis for $\mathbb{R}^3$, is $\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_2 + \mathbf{v}_3, \mathbf{v}_3 + \mathbf{v}_1\}$ necessarily a basis? Prove or give a counterexample.
4. (**Mechanical**) Let $\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$. Find the change-of-basis matrix $P_{\mathcal{B} \leftarrow \mathbf{B}}$ where $\mathbf{B}$ is the standard basis. Then find coordinates of $\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$ in basis $\mathcal{B}$.
5. (**Conceptual**) Explain why any set of 4 vectors in $\mathbb{R}^3$ cannot be linearly independent. Use the definition of basis.
6. (**Conceptual**) Suppose P is the change-of-basis matrix from $\mathcal{B}$ to $\mathcal{C}$. What is the relationship between P and the change-of-basis matrix from $\mathcal{C}$ to $\mathcal{B}$? Prove it.
7. (**Mechanical**) Determine if $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$ is a basis for $\mathbb{R}^3$. If so, find the coordinates of $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ in this basis.
8. (**Conceptual**) Can a basis for $\mathbb{R}^4$ contain the zero vector? Why or why not?

B Column Space and Row Space

1. (**Mechanical**) For $A = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}$, find a basis for the column space and state its dimension.
2. (**Conceptual**) Explain why the column space of an $m \times n$ matrix is a subspace of $\mathbb{R}^m$ and the row space is a subspace of $\mathbb{R}^n$.
3. (**Mechanical**) Find bases for the column space and row space of

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{pmatrix}.$$

4. (**Conceptual**) Is it possible that the column space of a 3×3 matrix is a line through the origin? If yes, give an example. If no, explain.
5. (**Mechanical**) For $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$, find a basis for the column space using the pivot columns from its reduced row echelon form.
6. (**Conceptual**) What is the relationship between $\text{rank}(A)$ and $\text{rank}(A^T)$? Prove it using column/row space dimensions.
7. (**Mechanical**) Let A be a 2×4 matrix. If the row space has dimension 2, what can you say about the column space? Give a concrete 2×4 matrix satisfying this and find bases for both spaces.
8. (**Conceptual**) Show that the row space and the null space are orthogonal complements in $\mathbb{R}^n$.
9. (**Mechanical**) For $A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, find bases for column space, row space, and null space. Verify the rank-nullity theorem.
10. (**Conceptual**) Prove that every vector in the column space can be written as $A\mathbf{x}$ for some $\mathbf{x}$. Conversely, if a vector can be written as $A\mathbf{x}$, it lies in the column space.

C Null Space

1. (**Mechanical**) Find the null space of $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. What is its dimension?
2. (**Conceptual**) Explain why the null space of a matrix is always a subspace.

3. (**Mechanical**) For $B = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix}$, find a basis for the null space. What is the nullity?
4. (**Conceptual**) Can the null space of a 3×3 matrix be the entire $\mathbb{R}^3$? If yes, what must the matrix be? If no, explain.
5. (**Mechanical**) Compute the null space of $C = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{pmatrix}$. Write the solution set in parametric vector form.
6. (**Conceptual**) If A is an $m \times n$ matrix with $\text{rank}(A) = n$, what is the null space? Why?
7. (**Mechanical**) Find all vectors in the null space of

$$D = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

(a 3×4 matrix). What is the nullity?

8. (**Conceptual**) Prove that if A is invertible, then the only solution to $A\mathbf{x} = 0$ is $\mathbf{x} = 0$. Relate to null space.
9. (**Mechanical**) For $E = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 1 \end{pmatrix}$, find the null space. Verify that null space vectors are orthogonal to each row of E .
10. (**Conceptual**) Suppose A is a 3×5 matrix with nullity 2. What is the rank? Can you determine the dimensions of the column space and row space? Explain.